Visvesvaraya National Institute of Technology (VNIT)Nagpur
Department of Electronics and Communication Engineering

ANALOG CIRCUIT DESIGN

PROJECT REPORT

TOPIC : RC PHASE SHIFT OSCILLATOR

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AIM : IMPLEMENTATION OF RC PHASE SHIFT OSCILLATOR USING OPERATIONAL AMPLIFIER.

APPARATUS : OPERATIONAL AMPLIFIER IC741, RESISTOR (33KΩ*2 , 3.6KΩ*3 , 2MΩ), CAPACITORS (0.1Uf*3), CONNECTING WIRES, PERV BOARD, IC HOLDER.

THEORY :

An **RC Phase Shift Oscillator** is a **linear oscillator** used to generate **pure sinusoidal waveforms** without requiring any external input signal. It is a **feedback oscillator**, which means it sustains oscillations using energy fed back from its output to its input.

Detailed Working Mechanism

1. **Op-Amp as the Active Element**:
   - The op-amp is used in **inverting amplifier mode**. It provides a **180° phase shift** and **amplifies** the signal.
   - Its **high input impedance** and **low output impedance** make it ideal for interfacing with the passive RC network.
2. **RC Phase Shift Network**:
   - Three RC sections are connected in series.
   - Each RC stage contributes a phase shift of **60°** at the oscillation frequency, adding up to **180°**.
   - The network acts as a **frequency-selective feedback path** — it allows a specific frequency (resonant frequency) to pass with correct phase.
3. **Barkhausen Criterion**: For sustained oscillations:
   - **Total phase shift** around the loop = 0 degrees OR 360 degrees
   - **Loop gain** $A\beta \geq 1$
     - A = amplifier gain
     - β = feedback factor (attenuation of RC network)

In this oscillator:

- $A = -Rf/R1$ (gain of inverting op-amp)

- B ≈ 1/29, so A ≥ 29

Using complex impedance analysis, each RC section contributes a frequency-dependent phase shift. At a specific frequency:

$$\omega = \frac{1}{RC\sqrt{6}} \quad \Rightarrow \quad f = \frac{1}{2\pi RC\sqrt{6}}$$

This is the frequency at which the total phase shift from the three RC sections becomes exactly 180°, matching the 180° from the op-amp — completing the feedback loop with zero net phase shift.

In practical, if gain is too high, the output amplitude grows uncontrollably; if too low, oscillations die out. To stabilize amplitude:

- **Nonlinear elements** (diodes, FETs) or **automatic gain control** circuits are used in some designs.
- In simple lab setups, we slightly overdesign the gain and allow natural circuit losses to settle the amplitude.

Advantages

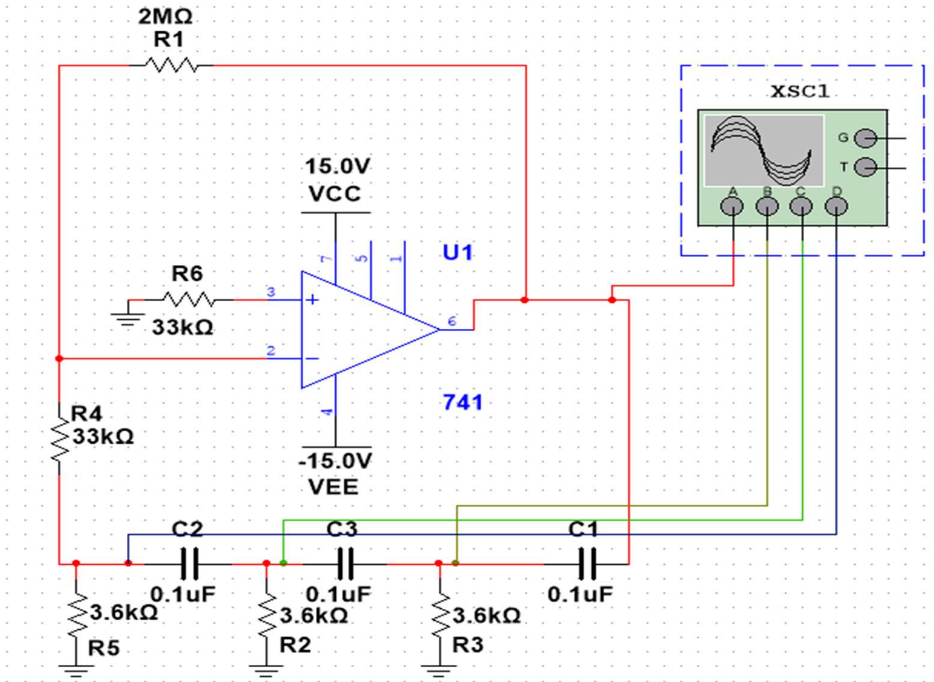
- Simple design using basic components.
- Pure sine wave output.
- Good frequency stability for low frequencies (audio range).

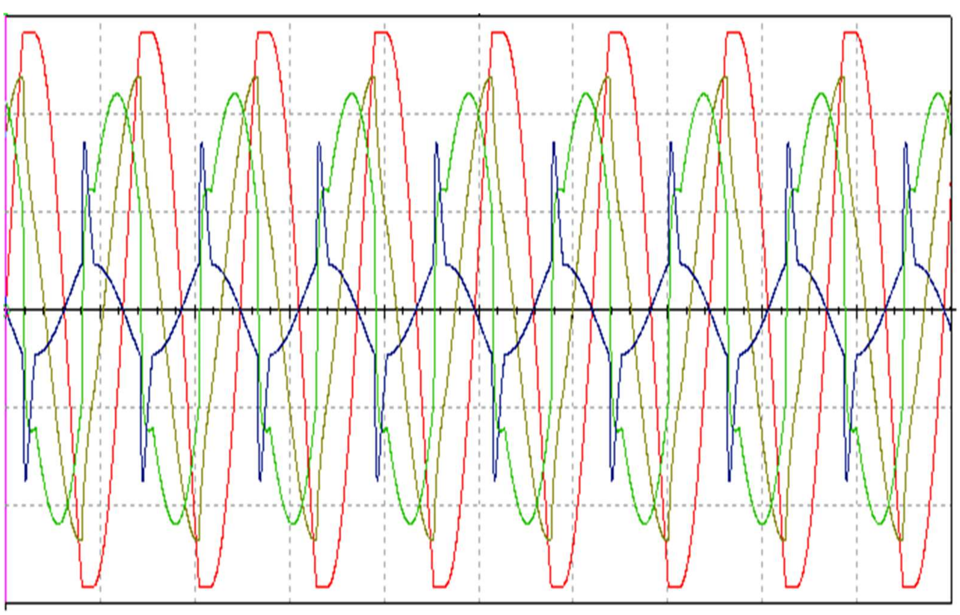
Limitations

- Not suitable for high frequencies (above few hundred kHz) due to RC network loading and op-amp limitations.
- Amplitude can drift if not properly stabilized.
- Requires large gain (≥ 29), which might cause distortion if not carefully handled.

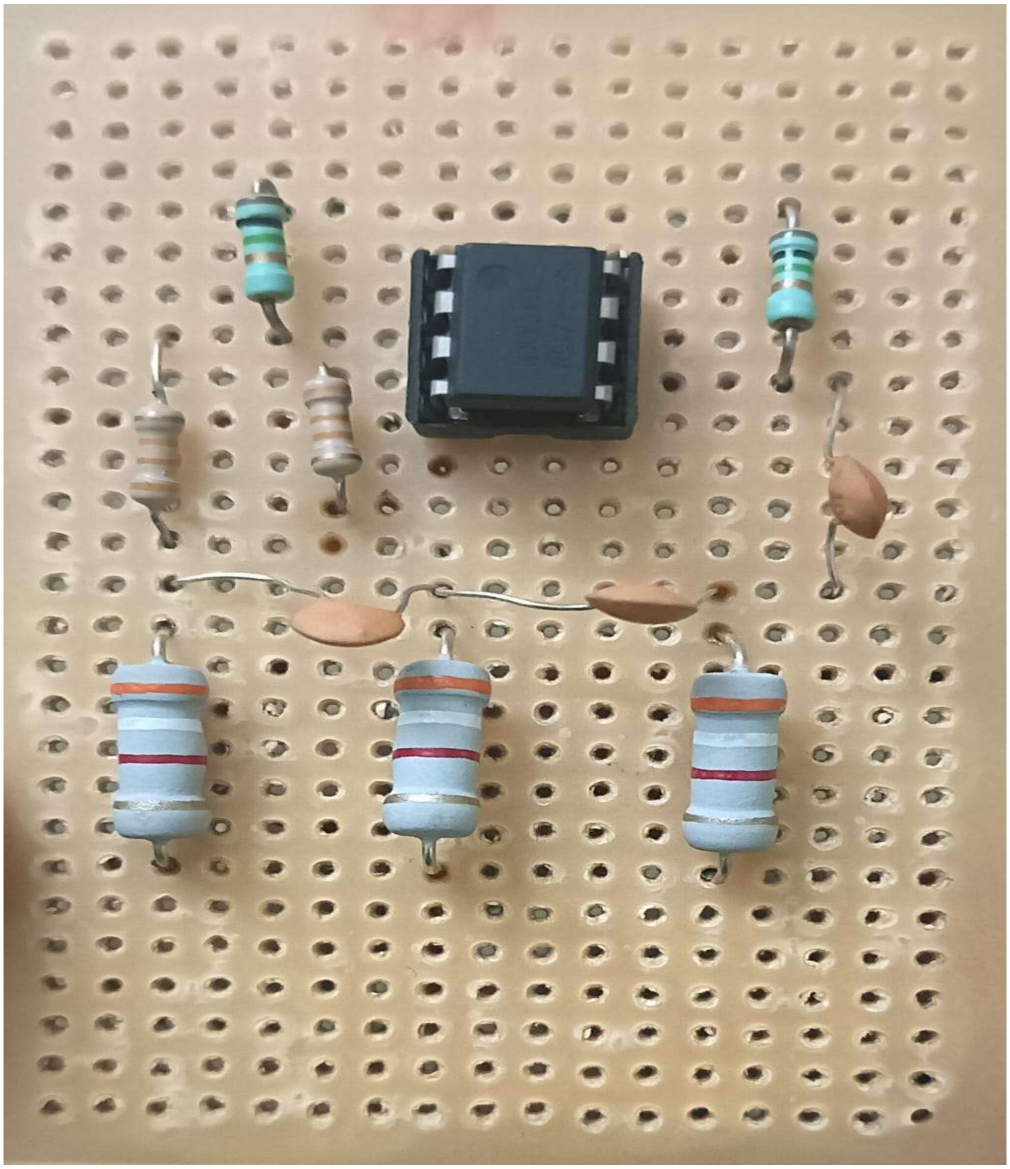
CIRCUIT DIAGRAM:

MULTISIM SIMULATION



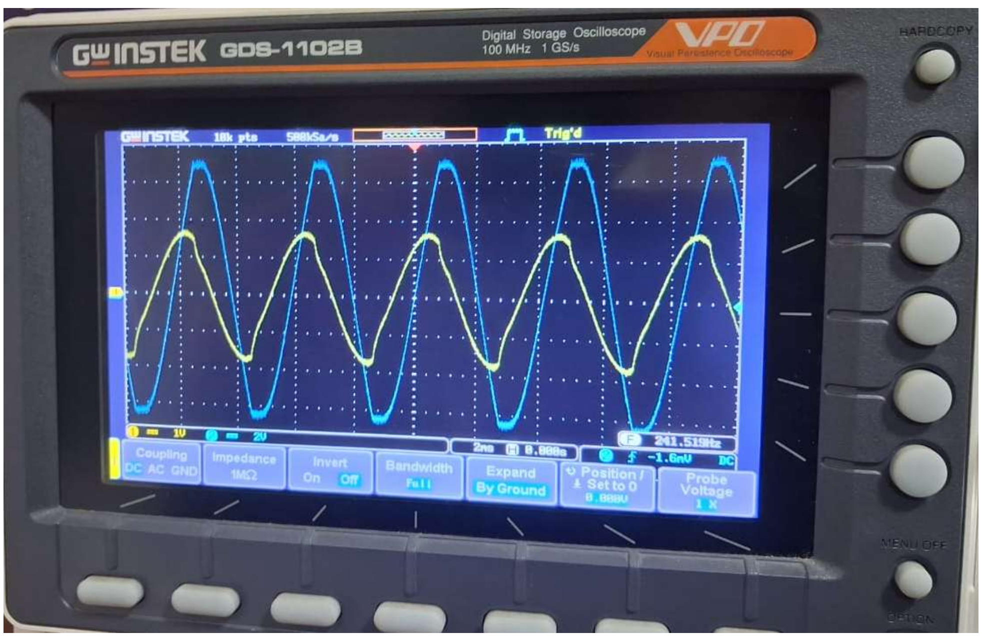


SOLDERED CIRCUIT ON PERF BOARD

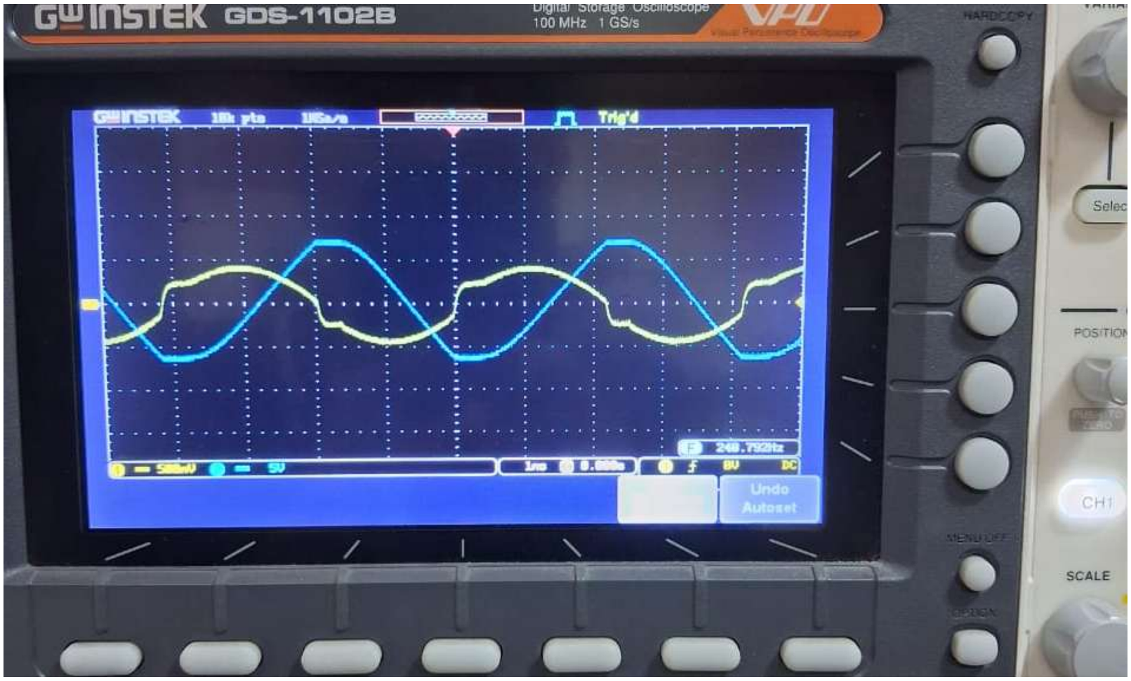


OBSERVATIONS : OUTPUT (DIGITAL STORAGE OSCILLOSCOPE)

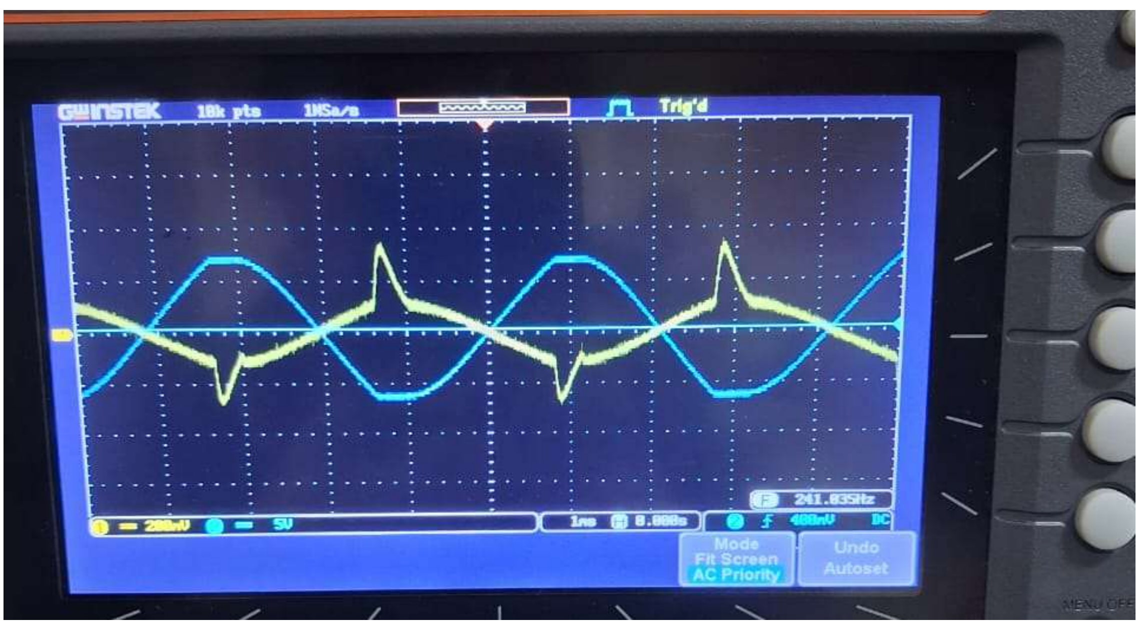
OUTPUT + INPUT 1R (60 DEGREE SHIFT)



OUTPUT + INPUT 2R (120 DEGREE SHIFT)



OUTPUT + INPUT 3R (180 DEGREE SHIFT)



CALCULATIONS:

Gain:

$$A = \frac{R_f}{R_{in}} = \frac{2 \times 10^6}{33 \times 10^3} = 60.61$$

Formula for Frequency:

$$f = \frac{1}{2\pi RC\sqrt{6}}$$

Given Values:

- $R = 3.6\,k\Omega$
- $C = 0.1\,\mu F$

Final Answer:

$$f = \frac{1}{2\pi \times 3.6 \times 10^3 \times 100 \times 10^{-9} \times \sqrt{6}} = 180.5\,\mathrm{Hz}$$

Gain of circuit : 60.61

Frequency of Oscillation : 180.5 Hz

Result:

The RC Phase Shift Oscillator was successfully constructed using a 741 operational amplifier. The oscillator produced a stable sine wave at 180.5 Hz of frequency.

Conclusion:

The experiment verified that a 741 op-amp can be used to build a phase shift oscillator. By using three RC stages to provide a total phase shift of 180°, and combining it with the 180° phase shift of the inverting op-amp, the Barkhausen criterion for sustained oscillations was satisfied. The observed frequency of oscillation closely matched the theoretical value.

Bibliography :

1. Gaikwad Ramakant A.
   Op-Amps and Linear Integrated Circuits.